

Quiz 3

1. All questions carry equal marks.
2. Justify your answers in a paragraph or less.

Question 1

Find an element $a = p + 7q + 49r \in \mathbb{Z}/\langle 7^3 \rangle$ with $p, q, r \in \{0, \dots, 6\}$ such that $a^2 = 8 \pmod{7^3}$.

Solution. Note that $8 = 1 \pmod{7}$. As we have proved in class the group of units $U_1(\mathbb{Z}_7)$ is 2-divisible. Hence, there is a 7-adic number $1 + 7q + 7^2r + \dots$ whose square is 8 in $\mathbb{Z}_7$. Reducing this modulo 7^3 will give the required element a .

We calculate

$$(1 + 7q + 49r)^2 = 1 + 2 \cdot (7q) + 49q^2 + 2 \cdot (49r) \pmod{7^3}$$

Since $8 = 1 + 7 \cdot 1 + 49 \cdot 0$, we need to solve

$$7 = 2 \cdot (7q) + 49q^2 + 2 \cdot (49r) \pmod{7^3}$$

First we solve $7 = 2 \cdot (7q) \pmod{7^2}$, or equivalently $1 = 2 \cdot q \pmod{7}$ to get $q = 4$.

We then need to solve

$$7 = 2 \cdot (7 \cdot 4) + 49 \cdot 4^2 + 2 \cdot (49r) \pmod{7^3}$$

or equivalently,

$$0 = 49 + 49 \cdot 4^2 + 2 \cdot (49r) \pmod{7^3}$$

or equivalently,

$$0 = 1 + 4^2 + 2r \pmod{7}$$

to get $r = 2$.

In other words, we have $a = 1 + 7 \cdot 4 + 49 \cdot 2 = 127$.

Question 2

Given that 2 is an element of multiplicative order 10 in $\mathbb{Z}/\langle 11 \rangle$, Find k in $\{0, \dots, 11\}$ such that if $a = 2 + 11k$, then a has multiplicative order 10 in $\mathbb{Z}/\langle 11^2 \rangle$.

Solution. We have seen in class that 2^{11^n} converges in $\mathbb{Z}_{11}$ to an element α of multiplicative order dividing $10 = 11 - 1$. Since 2^{11^n} is congruent to 2 modulo 11 and we are given that the latter has multiplicative order 10, it follows that the limit α has multiplicative order *exactly* 10 in $\mathbb{Z}_{11}$.

It follows that this limit α can be reduced modulo 11^2 to obtain a . In order to calculate a , we need m such that $v_{11}(2^{11^n} - 2^{11^m}) \geq 2$ for $n \geq m$. We can then take a to be the reduction of 2^{11^m} modulo 11^2 .

We have seen in class that if $p = q + 11^n r$ with $n \geq 1$, then $p^{11} = q^{11} + 11^{n+1} s$.

This shows that if $b = 2^{11} = 2 + 11r$, then $2^{11^n} - b$ is divisible by 11^2 for all $n \geq 1$. In other words, $a = b \pmod{11^2}$.

Thus, we need only calculate $b \pmod{11^2}$, or equivalently $r \pmod{11}$. We note

$$2^{11} = 2048 = 2 + 186 \times 11 = 2 + (10 + 16 \times 11) \times 11 = 2 + 10 \times 11 \pmod{11^2}$$

Hence $k = 10$ and $a = 2 + 11 \cdot 10 = 112$.